# Syed Mehrab

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## Professional Summary

AI-focused Software Engineer with expertise in building production-ready AI systems using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and backend microservices architecture. Proven track record in developing scalable chatbot platforms, deploying quantized LLMs, implementing gRPC-based distributed systems with GPU acceleration, fine-tuning multilingual text-to-speech models, optimizing inference pipelines with BentoML and vLLM, and containerizing AI services for cloud deployment. Strong foundation in Python, Django, FastAPI, RAG pipelines, vector databases, and AI automation workflows using n8n.

## Professional Experience

**AI Python Developer**
**Next Generation Innovations (NGI)** Karachi, Pakistan
February 2022 – Present

- Architected and deployed AI-powered chatbot systems leveraging Large Language Models and Retrieval-Augmented Generation for enterprise conversational AI applications
- Engineered gRPC microservices architecture enabling high-performance communication between distributed AI components with optimized latency and throughput
- Containerized AI services using Docker with GPU acceleration support, achieving production-ready deployment pipelines for text-to-speech and LLM inference workloads
- Fine-tuned multilingual TTS models for German and English using NVIDIA NeMo and Riva, improving speech naturalness and language-specific phonetic accuracy
- Deployed and optimized quantized Large Language Models including DeepSeek, Mistral-7B, and TinyLlama using BentoML and vLLM, reducing memory footprint while maintaining inference quality
- Initiated research and development of agentic AI systems for autonomous task execution and decision-making workflows
- Collaborated with cross-functional teams on infrastructure provisioning, performance optimization, and production support
- Contributed to Django-based backend development with focus on API reliability, database optimization, and system performance

**Trainee Python Developer**
**Next Generation Innovations (NGI)** Karachi, Pakistan
October 2021 – January 2022

- Developed Django-based web applications with emphasis on backend architecture, database design, and application performance optimization
- Implemented RESTful APIs for client-server communication and integrated third-party services

**Django Intern (Part-time)**
**Crop2x** Karachi, Pakistan
July 2021 – September 2021

- Built and maintained REST APIs using Django REST Framework with PostgreSQL database backend and Redis caching layer
- Debugged production issues and provided technical support using Jira for ticket management
- Implemented asynchronous task processing with Celery for background job execution and scheduled operations

**Web Developer Intern**
**Vivotic Solucion** Karachi, Pakistan
June 2021 – August 2021

- Developed front-end features for web applications using JavaScript and modern web development frameworks

- Maintained and enhanced existing web application codebases with focus on user interface responsiveness and cross-browser compatibility

## Selected Projects

**AI-Powered Virtual Support Agent (heronOS)**

Customizable LLM and RAG-based virtual agent providing end-to-end online customer support with video avatar capabilities.

- Led development of LLM- and RAG-powered conversational systems for advanced, context-aware interactions
- Built gRPC microservices with streaming support for efficient communication between distributed AI components
- Containerized AI services with Docker and GPU acceleration for scalable production deployments
- Fine-tuned multilingual TTS models (English & German) using NVIDIA NeMo/Riva for enhanced naturalness and accuracy
- Optimized quantized LLMs (DeepSeek, Mistral-7B, TinyLlama) using BentoML and vLLM for reduced memory footprint and high inference performance

**Fintech Savings & Investment Platform (Confidential Fintech Solution)**

Interactive platform enabling small-amount savings, multi-investment management, and real-time portfolio tracking via dynamic dashboards.

- Developed secure Django backend APIs to handle savings, investments, transactions, and financial data protection
- Built graph-based APIs for displaying portfolio statistics and delivering clear financial insights
- Integrated push notifications and SMS services for real-time user engagement and updates

**Telehealth Platform (Health-at-Hand)**

Accessible health services platform with secure data handling, user-focused features, and healthcare interoperability standards.

- Engineered Django backend services for secure management of health-related data and transactions
- Implemented interactive dashboards and notification systems to enhance user experience and engagement
- Integrated HL7 FHIR-compliant RESTful APIs to enable seamless interoperability with external systems (EHRs, labs, monitoring devices), supporting standardized clinical data exchange and regulatory compliance

**Automated Lead Generation & Booking System (n8n Workflow)**

Intelligent automation integrating RAG-powered chatbot with calendar booking and inbound call handling (e.g., for gyms/fitness businesses).

- Designed RAG-enhanced LLM chatbot for natural, context-aware conversations
- Built automated calendar booking logic integrated with inbound call systems
- Created lead capture, qualification, and generation workflows for business clients

## Technical Skills

- **Languages:** Python, JavaScript, SQL
- **Backend & Frameworks:** Django, FastAPI, Django REST Framework
- **AI & LLM:** Large Language Models (LLMs), Hugging Face Transformers, BentoML, vLLM, Model Quantization, RAG, Vector Databases, Embedding Models
- **TTS & Audio:** Multilingual TTS, NVIDIA NeMo, NVIDIA Riva, Speech Synthesis, Audio Processing, Fine-tuning
- **Models:** DeepSeek, Mistral-7B, TinyLlama
- **Infrastructure & Deployment:** Docker, GPU Acceleration, gRPC, Microservices, REST APIs, Git, CI/CD Pipelines
- **Databases & Caching:** PostgreSQL, Redis
- **Task Processing:** Celery, Background Jobs

- **AI Automation:** n8n, Agentic AI Systems, Workflow Orchestration
- **Frontend:** Next.js, Vercel AI SDK, JavaScript, Responsive Design

## Education

**Bachelor of Science in Software Engineering**
Iqra University, Karachi, Pakistan — June 2021
CGPA: 3.01

**HSC (Intermediate), Science**
Malir Degree Government Science College, Karachi, Pakistan — May 2015

## Certifications

- Artificial Intelligence Certificate (AIC) – Presidential Initiative for Artificial Intelligence & Computing (PIAIC) December 2022
- Mobile App Development and Web Graphics – Institute of Business Administration (IBA) March 2021